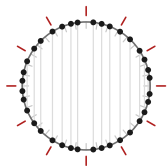
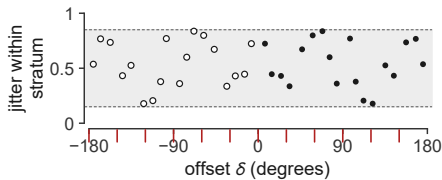


a Fixed nonuniform offset design

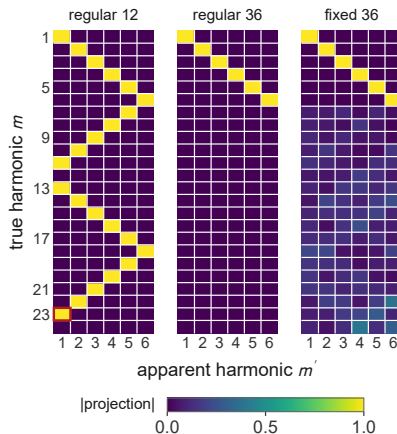


36 fixed offsets, mirror pairs joined
regular 12-point design (outer ticks)



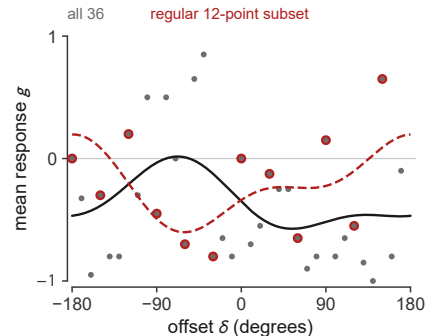
36 offsets = 18 exact mirror pairs; no anchor at $\delta = 0$.
Stratified jitter in $\pi/18$ strata, seed 2026102300.
Gaps 6.08–14.47°; jitter within [0.15, 0.85].

b Aliasing onto low-order modes



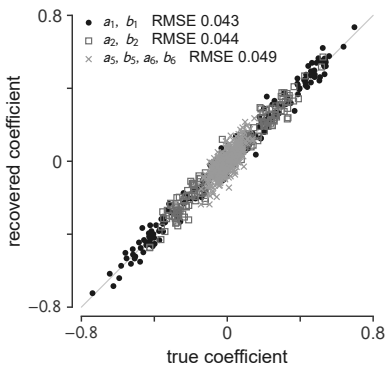
Regular 12-point sampling maps $m = 23$ onto $m' = 1$ with unit gain (red cell).
That harmonic carries amplitude 0.242 in the narrow field; the fixed design
leaks 0.075 of it. A regular 36-point grid is clean to $m = 29$, not beyond.

c Measured sign change



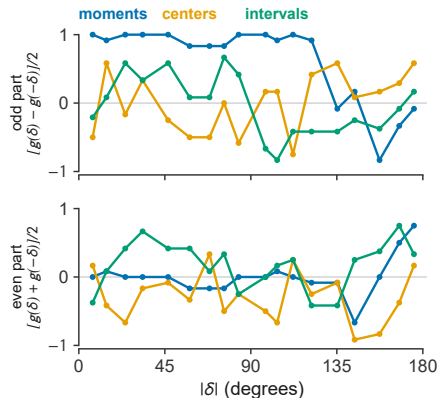
centers, narrow field $\kappa = 12$; 40 per offset.
36-point grid, black fit: $a_1 = -0.242$
12-point subset, red fit: $a_1 = +0.134$
other subset phases: $-0.459, -0.400$

d Recovery on measured offsets



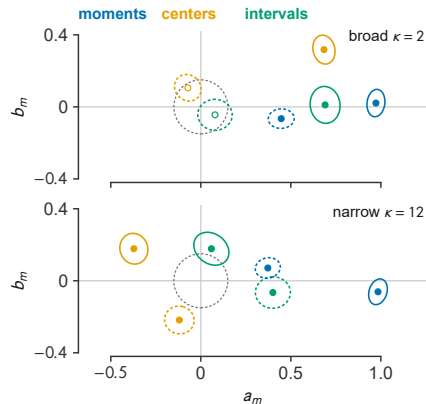
90 mixtures, 24 responses per offset, order-6 fit
on the 36 fixed offsets, with an out-of-band $m = 23$
term at the measured amplitude 0.242.

e Mirror-pair balance



Unimodal $\kappa = 9$, block 1, 18 mirror pairs.
The design supplies both halves of every pair; it
does not make the response odd. mean [even part]:
moments 0.153; centers 0.387; intervals 0.333.

f Bootstrap stability of C_m



Block 1, 95% multinomial bootstrap. Solid
ellipse C_1 , dashed C_2 ; open centre marks
a suppressed phase ($R_m < 0.15$, dotted circle,
or ellipse covering the origin).